Design and Sizing of a Renewable Microgrid for 24-Hour EV Fleet Charging Using Xendee Microgrid Toolkit and Sensor Data

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Introduction

The rapid growth of electric vehicle (EV) adoption, particularly in commercial fleet applications, has raised critical challenges for grid reliability, energy infrastructure planning and sustainable operations. Earlier projects in the M3TFluiD lab focused on assessing solar and wind resources support EV charging; this project builds on those findings by transitioning from resource evaluation to system level microgrid design using known renewable energy potential at the Valley View, Ohio site based on prior sensor deployments this study aims to design in size a hybrid wind solar micro grid for 24-hour EV fleet charging.

Abstract

This project leverages the Xendee techno-economic platform and ESAP (Energy Systems Analysis Platform) to evaluate system configurations that meet demand reliably while remaining cost-effective across the project lifespan. We explored how energy storage integration, component sizing and operational strategies affect system performance and cost under varying load profiles. This work directly supports the broader goal of creating scalable, replicable microgrids that alleviate stress on centralized grids, enhancing delivery fleet flexibility and reducing operational emissions. This research experience is directly translated to real world tools and workflows used in microgrid planning and improved skills in system modeling, scenario analysis and sustainability planning. The outcome will contribute to an evolving body of work on decentralized solutions for transportation electrification and urban energy resilience.

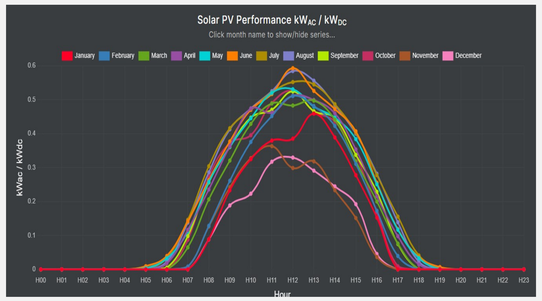
Methods

The first step was to determine the assumptions and known values to work with. In coordination with a Schneider Electric representative, I received information from one of their projects that I could use as assumptions for mine. These numbers were based on a standard 8-hour 9am to 5pm shift. ICE represents the typical internal combustion engine delivery vehicle. These values allowed me to determine the electrical load required to charge all 10 vehicles and have the availability to charge them 24-hour if necessary. The load was 102kW per delivery truck per active day.

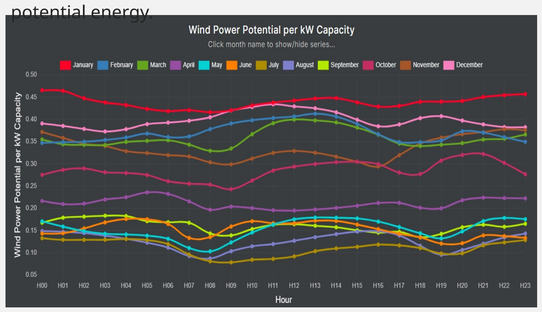
Methods

Years in Service	5
Annual Miles	45,000
Days per year	260
Shifts Per Day	1
Idling Hours Per 8-Hour Shift (hours)	1
Competitor ICE Miles per Gallon (mpg)	17
Competitor ICE Idling Consumption per hour (gallon)	0.5
EV Miles per kWh (mpk)	1.7
EV Idling Consumption per hour (kWh)	1
Price per Gallon ($)	$2.50
Price per kWh ($)	$0.14
ICE Maintenance Cost per Mile ($)	$0.101
BEV Maintenance Cost per Mile ($)	$0.061

After receiving assumption values and determining the loads required per day I collaborated with other students in the lab to find the wind and solar data for that area to determine how much energy could be created.



Xendee creates visuals for the CSV data you input and organizes by hour of day and month of year. Each point in the graphs represents the average of a specific hour of the day. The solar graph is direct energy while the wind graph is the potential energy.

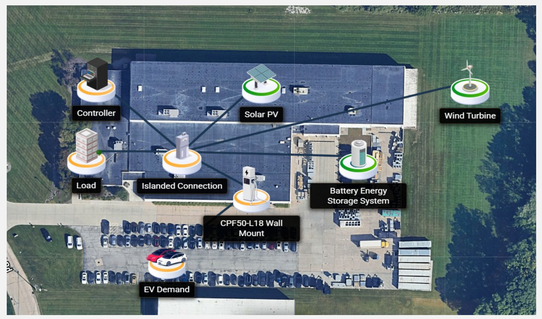


After receiving assumption values, I found the wind and solar data I used the Website called PVWatts. The website PVWatts allows you to calculate solar power capacity based on a drawn shape over the Google Maps satellite image. After collecting the information, I was able to experiment with the platform Xendee.

Findings



Xendee allows for both microgrid design and optimization, so I was able to insert the data found and then decide how I wanted the grid to be optimized. The two tracks I decided to focus on were maximizing the roof solar energy and minimizing the number of overall units to be bought.



The first optimization was designed to reduce the number of purchases necessary to complete the grid. The result of that was purchasing more wind power compared to solar. This optimization purchased 4 100kW turbines to cover a majority of the load with the solar energy being placed into storage. This grid has 696 kW of storage connected to it because there is not the full 380kW potential solar energy connected to the system.

Type	Total New Capacity	Technology (New Capacity)	Init. Equip. Cost
	34.3 kW	Solar PV (34.3 kW)	71 k$
	696 kWh	Battery Energy Storage System (696 kWh)	268 k$
	400 kW	Wind Turbine (400 kW) (4 units)	1,600 k$
	120 kW	CPF50-L18 Wall Mount - CPF50-L18 Wall Mount (120 kW) (10 units)	19 k$
			Total Cost: 1,958 k$

Findings

Type	Total New Capacity	Technology (New Capacity)	Init. Equip. Cost
	380 kW	Solar PV (380 kW)	784 k$
	1.2 MWh	Battery Energy Storage System (1.2 MWh)	463 k$
	100 kW	Wind Turbine (100 kW) (1 unit)	400 k$
	120 kW	CPF50-L18 Wall Mount - CPF50-L18 Wall Mount (120 kW) (10 units)	19 k$
			Total Cost: 1,666 k$

This second model creates almost double the storage capacity and uses less wind energy. This is because the fleet charges overnight when solar energy is no longer being produced. Requiring high solar and high battery storage allows for more passive energy collection during the day. If anything were to happen to the battery system, the turbine would support the grid.

Limitations

Some limitations of this project would be current knowledge and software experience. As someone new to the energy field, there is a good learning curve when discovering how these alternative energies work and how to use them together. Learning the subject while also learning a new software created large amounts of troubleshooting.

Discussion

The second model is what I would recommend Schneider use as there is potential to sell the extra stored energy created, less construction and expansion required, and overall capital expenditures are less. Something to follow up on for this project is learning more about the long-term financial breakdown of each model. Currently there are the initial purchasing costs that do not account for external variables. These models were also created to be islanded (disconnected from the main grid) so looking at connection points to allow for energy to be sold or redirected may save Schneider money in other areas.

Acknowledgements

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